

4. Phases of the Moon

1. Draw a picture of the earth, orbited by the moon
 - A. Draw the moon at each of the eight phase positions
 - B. Show the Sun's rays coming from the left
 - C. Shade the illuminated portion of the each of the eight moons, and the illuminated half of the Earth
 - D. Draw a line showing the half of the moon closest to the Earth at each phase position
 - E. Indicate the observers time from noon to midnight in the proper position
2. Why do we see phases of the Moon?
3. If you observe the Moon in the figure below, what phase will it have?



4. If the third quarter Moon rose nine hours ago, what time of the day or night is it?
5. If you observe the Moon in the figure below almost directly overhead in the sky, what time of the day or night is it?



6. If the Moon is moving up through the sky (well above the eastern horizon, not yet overhead) in the sky in the evening (9pm), what phase will it have?
7. How long, in days, does it take the Moon to shift from a first quarter phase to a new moon phase?
8. This figure shows the location and phase of the Moon as we look from our position in the northern hemisphere toward the southern sky, with the horizon stretching from east (on the left) to west (on the right). What time of the night or day is shown?

